SIMULATOR
EXAMINATION SCENARIO GUIDE

SCENARIO TITLE:	15-01 NRC ESG-3
SCENARIO NUMBER:	15-01 NRC ESG-3
EFFECTIVE DATE:	See Approval Dates
EXPECTED DURATION:	60 minutes
REVISION NUMBER:	00
PROGRAM:	☐ **L.O. REQUAL**
	X **INITIAL LICENSE**
	☐ **STA**
	☐ **OTHER** ____________

Revision Summary
New issue for 15-01 ILOT NRC exam

PREPARED BY:	G Gauding **Lead Regulatory Exam Author**	8-11-16 **Date**
APPROVED BY:	M. Fla **Operations Training Manager**	10/12/16 **Date**
APPROVED BY:	J. Fleming **Facility Representative**	10/18/16 **Date**

SCAN OF SIGNED SCENARIO COVER SHEET

I. OBJECTIVES

A. Given the order or indications of a pressurizer control system malfunction, perform actions as the nuclear control operator to RESPOND to the malfunction in accordance with S1/S2.OP-S2.OP-AB.PZR-0001.

B. Given indication of a pressurizer control system malfunction, DIRECT the response to the malfunction in accordance with S2.OP-AB.PZR-0001.

C. Given indication of a dropped control rod perform actions as the nuclear control operator to RESPOND to the malfunction in accordance with S2.OP-AB.ROD-0002.

D. Given indication of a dropped control rod DIRECT the response to the malfunction in accordance with S2.OP-AB.ROD-0002.

E. Given the order or indications of a loss of secondary coolant, perform actions as the nuclear control operator to RESPOND to the coolant loss in accordance with EOP-LOSC-1.

F. Given indication of a loss of secondary coolant, DIRECT the response to the loss of secondary coolant, in accordance with EOP-LOSC-1.

G. Given the order or indications of a reactor trip, perform actions as the nuclear control operator to RESPOND to the reactor trip in accordance with 2-EOP-TRIP-1.

H. Given indication of a reactor trip, DIRECT the response to the reactor trip in accordance with 2-EOP-TRIP-1.

I. Given the order or indications of a safety injection, perform actions as the nuclear control operator to RESPOND to the safety injection in accordance with the approved station procedures.

J. Given indication of a safety injection, DIRECT the response to the safety injection in accordance with the approved station procedures.

K. Given the order or indications of a multiple steam generator depressurization, perform actions as the nuclear control operator to RESPOND to the generator depressurization in accordance with EOP-LOSC-2.

L. Given indication of a multiple steam generator depressurization, DIRECT the response to the generator depressurization in accordance with EOP-LOSC-2.

II. MAJOR EVENTS

A. Power Ascension

B. Controlling PZR Pressure Instrument fails high

C. Dropped control rod

D. Steam Leak/Rupture in containment

III. SCENARIO SUMMARY

A. The crew will take the watch with the unit stable at $1x10^{-8}$ Amps during a plant startup, BOL. 21 SGFP is in service and 22 SGFP is O/S. Steam dumps are in MS Pressure Control – Auto @ 1,000 psig. 25 CFCU is C/T for maintenance. 23 charging pump is C/T for emergent maintenance. The crew will be instructed to raise power to 2%.

B. The crew will initiate a power ascension to 2% using Main Steam Dumps and control rods.

C. After the power ascension is started the controlling PZR Pressure instrument will fail high, deenergizing PZR heaters and causing both PZR Spray valves to open fully. The crew will respond IAW S2.OP-AB.PZR-0001, Pressurizer Pressure Malfunction to establish manual control of PZR pressure control and restore pressure to normal, swap to an operable channel, and remove the failed channel from service. The CRS will identify applicable LCO's.

D. After the Pressurizer Pressure control system failure has been addressed, a control rod will drop fully into the core. The crew will respond IAW S2.OP-AB.ROD-0002, Dropped Rod. The crew will identify the Rx is subcritical, and initiate control rod insertion to fully insert all Control and Shutdown rods. The CRS will identify applicable LCO's.

E. During the rod insertion, a steam leak in containment will occur on 21 SG. The crew will diagnose changing containment conditions, and determine a Rx trip is required and MSLI will be required in an attempt to isolate the steam leak.

F. The crew will trip the Rx. The steamline will rupture. MSLI (auto and manual) will fail to shut any MSIV. The crew will initiate Safety Injection when it is identified that the source of the steam leak is not isolated.

G. The crew will perform plant stabilization actions in EOP-TRIP-1, Reactor Trip or Safety Injection. With all SGs faulted, the crew will maintain AFW flow to each SG. 21 AFW pump will trip during TRIP-1 performance. Containment Spray pumps will not automatically start when demanded on hi-hi containment pressure, and the crew will manually start at least one CS pump (CT-1). The crew will transition to EOP-LOSC-1, Loss of Secondary Coolant.

H. The crew will transition to EOP-LOSC-2, Multiple Steam Generator Depressurization with all SG's faulted.

I. The crew will identify the RCS cooldown rate of >100°F / hr, and control AFW flow to minimize RCS cooldown (CT-2). The crew will reset Safeguards. When evaluating SI Flow Reduction Criteria, local operators will successfully shut a MSIV. The crew will recognize rising pressure in that SG, and transition to LOSC-1.

J. The scenario will be terminated upon transition to LOSC-1.

IV. INITIAL CONDITIONS

____ IC-233

PREP FOR TRAINING (i.e. computer setpoints, procedures, bezel covers ,tagged equipment)

Initial		Description
____	1	VC1and VC4 C/T
____	2	RCPs (SELF CHECK)
____	3	RTBs (SELF CHECK)
____	4	MS167s (SELF CHECK)
____	5	500 KV SWYD (SELF CHECK)
____	6	SGFP Trip (SELF CHECK)
____	7	23 CV PP (SELF CHECK)
____	8	**21 Charging pump in service**
____	9	**25 CFCU C/T**
___	10	**23 Charging pump C/T**
___	11	**IOP-3 open and complete up to Section 4.3, Power Operation.**
___	12	Complete Attachment 2 “Simulator Ready-for-Training/Examination Checklist.”

Note: Tables with blue headings may be populated by external program, do not change column name without consulting Simulator Support group

EVENT TRIGGERS:

Initial	ET #	Description	
	1	EVENT ACTION:	MONP254 <10 // CONT ROD BANK C < 10 (RX TRIP)
		COMMAND:	
		PURPOSE:	<update as needed>

MALFUNCTIONS:

SELF-CHECK	Description	Delay Time	Initial Value	Ramp Time	Trigger	Severity
______ 01	PR0016A PZR PRESS CH I (PT455) FAILS H/L	N/A	N/A	N/A	RT-1	2500
______ 02	RD0267 ANY ROD(S) INADVERTENTLY DROPS	N/A	N/A	N/A	RT-3	3
______ 03	MS0090Ar 21 Main Steam Line Leak Inside Cont AFTER orifice	N/A	N/A	N/A	RT-5	2
______ 04	RP0279A AUTO MSLIS FAILS TO ACT, TRN A	N/A	N/A	N/A	N/A	
______ 05	RP0279B AUTO MSLIS FAILS TO ACT, TRN B	N/A	N/A	N/A	N/A	
______ 06	MS0092E 21MS167 FAILS OPEN	N/A	N/A	N/A	N/A	
______ 07	MS0092F 22MS167 FAILS OPEN	N/A	N/A	N/A	N/A	
______ 08	MS0092G 23MS167 FAILS OPEN	N/A	N/A	N/A	N/A	
______ 09	MS0092H 24MS167 FAILS OPEN	N/A	N/A	N/A	N/A	
______ 10	RP0274A AUTO SI FAILS TO ACT, TRN A	N/A	N/A	N/A	N/A	
______ 11	RP0274B AUTO SI FAILS TO ACT, TRN B	N/A	N/A	N/A	N/A	
______ 12	AF0181A 21 AUX FEEDWATER PUMP TRIP	00:05:00	N/A	N/A	ET-1	
______ 13	RP318L1 21 CS Pump Fails to Start on SEC	N/A	N/A	N/A	N/A	
______ 14	RP318L2 22 CS Pump Fails to Start on SEC	N/A	N/A	N/A	N/A	

REMOTES:

SELF-CHECK	Description	Delay Time	Initial Value	Ramp Time	Trigger	Condition
______ 01	CV52D 23 CHG PUMP BKR CONTROL POWER	N/A	N/A	N/A	N/A	OFF
______ 02	PR34D PORV STOP VALVE 2PR6 TAGGED	N/A	N/A	N/A	RT-7	TAGGED
______ 03	SP01A1 Prevailing Wind Speed	N/A	N/A	N/A	N/A	10
______ 04	SP01A2 Prevailing Wind Direction	N/A	N/A	N/A	N/A	160
______ 05	CT195-1D 25 CFCU BKR #1 High Speed 125VDC	N/A	N/A	N/A	N/A	OFF
______ 06	CT195-2D 25 CFCU BKR #2 High Speed 125VDC	N/A	N/A	N/A	N/A	OFF

_______ 07	CT195-3D 25 CFCU BKR #3 Low Speed 125VDC	N/A	N/A	N/A	N/A	OFF

OVERRIDES:

SELF-CHECK	**Description**	**Delay Time**	**Initial Value**	**Ramp Time**	**Trigger**	**Condition/ Severity**
_______ 01	CI11 C LO QCI11HR2 TURNING GEAR-ENGAGED	N/A	N/A	N/A	N/A	ON
_______ 02	CI11 E LO QCI11PR2 TURNING GEAR-STARTED	N/A	N/A	N/A	N/A	ON
_______ 03	CI11 F LO QCI11TG2 TURNING GEAR-STOPPED	N/A	N/A	N/A	N/A	OFF
_______ 04	CI11 D LO QCI11LG2 TURNING GEAR-DISENGAGED	N/A	N/A	N/A	N/A	OFF

OTHER CONDITIONS:

	Description

_____ 1. None

V. SEQUENCE OF EVENTS

A. State shift job assignments.

B. Hold a shift briefing, detailing instruction to the shift: (provide crew members a copy of the shift turnover sheet).

C. Inform the crew "The simulator is running. You may commence panel walkdowns at this time. CRS please inform me when your crew is ready to assume the shift".

D. Allow sufficient time for panel walk-downs. When informed by the CRS that the crew is ready to assume the shift, ensure the simulator is cleared of unauthorized personnel.

Evaluator/Instructor Activity	Expected Plant/Student Response	SBT LOG	Comment
1. Power ascension			
	CRS directs power ascension using Main Steam Dumps and control rods.		
	PO raises steam dump demand.		
	RO adjusts control rods to maintain Tave on program.		
	RO announces when POAH is observed.		
Continue to next event on direction from Lead evaluator.			
2. Controlling PZR pressure instrument fails HIGH			
Simulator Operator: Insert <u>**RT-1**</u> on direction from Lead Evaluator. **MALF: MALF: PR0016A PZR Press CH I (PT-455) fails H/L Final Value: 2500**			
	RO announces unexpected OHA alarms D-8 RC Press HI, and E-42 2PR1 ½ Trip.		
	RO determines actual pressure is not high and reports spray valves open, and recommends placing Master Pressure Controller in manual.		
	CRS directs RO to place Master Pressure Controller in manual.		

Evaluator/Instructor Activity	Expected Plant/Student Response	SBT LOG	Comment
	RO takes manual control of Master Pressure controller, and lowers demand (increase pressure) to close sprays.		
	CRS enters S2.OP-AB.PZR-001, Pressurizer Pressure Malfunction.		
	CRS directs initiation of AB.PZR CAS.		
	CRS gives band for control of PZR pressure.		
	RO identifies PZR Press Channel I failed high.		
	RO selects Channel III for control.		
	RO matches master pressure controller demand to current pressure and returns Master Pressure Controller to AUTO.		
	RO shuts 2PR6.		
Note: The requirement to shut 2PR6 and deenergize it is within one hour of the PZR Pressure Channel I instrument failure IAW Tech Specs.			
	CRS directs WCC to remove power from 2PR6.		
Simulator Operator: Insert RT-7 2 minutes after being directed to remove power from 2PR6. **REMOTE: PR34D PORV Stop**			

Evaluator/Instructor Activity	Expected Plant/Student Response	SBT LOG	Comment
Valve 2PR6 tagged			
	CRS directs PO to initiate removing failed channel from service and contact I&C for support IAW S2.OP-SO.RPS-003.		
	PO checks that tripping associated bistables will not result in an ESF or RPS actuation.		
Note: Crew may wait for I&C response before performing any actions in S2.OP-SO.RPS-0003.			
	PO verifies that Channel III is selected for Master Pressure Control.		
	PO selects PZR Pressure recorder to channel other than I.		
	CRS enters TSAS 3.3.1.1 Action 6, 3.3.2.1.b Action 19*, 3.4.5.b, and 3.2.5.		
Proceed to next event on direction from Lead Evaluator.			
3. Dropped control rod			
Simulator Operator: Insert **RT-3** on direction from Lead Evaluator. **MALF: RD0064 ANY ROD INADVERTENTLY DROPS Severity: 3**			
	RO reports unexpected OHA E-48 ROD		

Evaluator/Instructor Activity	Expected Plant/Student Response	SBT LOG	Comment
	BOTTOM, and reports Shutdown Bank rod 1SA3 rod bottom light is illuminated.		
	CRS enters S2.OP-AB.ROD-0002, Dropped Rod.		
	RO reports control rods are in manual.		
	PO reports Main Turbine is S/D.		
	RO reports the reactor is subcritical as a result of the dropped rod.		
	CRS directs RO to insert all control and shutdown bank rods.		
	RO initiates insertion of control rods in manual.		
Simulator Operator: Once rod insertion has commenced, insert <u>**RT-5**</u> on direction from Lead Evaluator **MALF: MS0090Ar** **Severity: 2**			
4. 21 SG steam leak / rupture in containment			
	PO announces OHA C-38, CFCU LK DET HI as unexpected, and shortly after, OHA C-30, CFCU LK DET HI-HI as well.		
	PO refers to ARP for OHA C-38 and C-30.		
	PO reports indications of a steam leak inside		

Evaluator/Instructor Activity	Expected Plant/Student Response	SBT LOG	Comment
	containment including: - Containment pressure rising - Tavg slowly lowering - Rising steam flows on all loops - Steam dump demand lowering		
	CRS enters S2.OP-AB.STM-0001, Excessive Steam Flow.		
	CRS directs monitoring of AB.STM CAS.		
	CRS notifies Emergency Services of the steam leak.		
	PO reports that the Main Turbine is not latched.		
	PO reports no indication of MS10 or steam dump valves malfunction.		
	RO reports no indication of a SG Safety Valve partially open or leaking.		
	RO reports Rx power and RCS temperature are NOT stable.		
	CRS directs the RO to trip the Rx.		
Note: A MSLI will be required to attempt to isolate the steam leak in containment. The CRS can use the CAS of AB.STM for direction, or will use Step 11 in TRIP-1 when containment pressure exceeds 15 psig. Auto and Manual MSLI are			

Evaluator/Instructor Activity	Expected Plant/Student Response	SBT LOG	Comment
failed and will not actuate.			
Simulator Operator: After the Rx is tripped, then **MODIFY MALF MS0090Ar** to 100 with a 5 minute ramp.			
	RO trips the Rx and confirms the Rx trip.		
Simulator Operator: Ensure **ET-1** is true upon the Rx trip. This trips 21 AFW pump after a 5 minute delay.			
5. All MSIVs fail to shut			
	IF using AB.STM CAS, then RO will attempt a MSLI, and reports it did not actuate.		
	IF using AB.STM CAS, RO reports the steam leak is NOT isolated.		
	IF using AB.STM CAS, RO initiates SI and begins performing TRIP-1 immediate actions.		
	RO continues Immediate Actions of TRIP-1: - Reports the Main Turbine is tripped. - Reports all 4KV vital buses are energized. - Reports a demand for Safety Injection is present and Safety Injection has NOT actuated (or reports Safety injection was manually initiated as described in		

Evaluator/Instructor Activity	Expected Plant/Student Response	SBT LOG	Comment
	previous step).		
	RO initiates Safety Injection (if not previously performed.)		
Note: With **all** SGs faulted, AFW flow should not be isolated to **any** SG.			
	PO reports SEC loading is not complete.		
	PO reports all available equipment started.		
	PO reports 21 and 22 AFW pumps are running, and if not previously performed, receives permission to throttle total AFW flow to no less than 22E4 lbm/hr.		
	PO reports valve groups in Table B are in Safeguards position.		
	RO reports 21 and 22CA330 are shut.		
6. 21 AFW pump trip			
	PO reports when 21 AFW pump trips, and adjusts AFW flow if required.		
7. Containment Spray pumps fail to start			
	RO reports containment pressure has not remained less than 15 psig.		
	RO initiates Phase B and Spray actuation.		

Evaluator/Instructor Activity	Expected Plant/Student Response	SBT LOG	Comment
	RO reports neither Containment Spray pump has started.		
	RO blocks 2A and 2C SECs.		
	PO resets 2A and 2C SEC's.		
	RO starts 21 and 22 CS pumps.		
CT#1 (CT-3) Manually start at least one Containment Spray pump before a red path challenge develops on the containment CSF. **SAT_____ UNSAT_____**			
	RO reports MSLI has failed to close any MSIV.		
	RO stops all RCPs.		
	RO reports MSLI has failed to close any MSIV.		
	CRS dispatches operators to place valves from Table D (locally close MSIVs) in safeguards position if not previously performed.		
	PO reports all 4KV vital buses are energized.		
	RO reports control room ventilation is in Accident Pressurized mode.		

Evaluator/Instructor Activity	Expected Plant/Student Response	SBT LOG	Comment
	RO reports 2 switchgear supply and 1 exhaust fan are running.		
	RO reports 2 CCW pumps running.		
	RO reports ECCS flow as expected for current RCS pressure.		
	PO maintains total AFW flow greater than 22E4 lbm/hr until at least one SG NR level is >15% (adverse), then maintains SG NR level 19-33%.		
	RO reports no RCPs are running.		
	RO reports RCS Tcolds are dropping.		
	PO reports no dumping steam.		
	RO reports MSLI failed to isolate any MSIV.		
	RO reports both RTBs are open.		
	RO reports both PZR PORVs are closed.		
	RO reports 2PR6 is shut, and 2PR7 is open.		
	CRS dispatches operator to restore power to 2PR6 if deenergized.		
	RO reports no RCPs are running.		
	RO maintains seal injection flow to all RCPs.		

Evaluator/Instructor Activity	Expected Plant/Student Response	SBT LOG	Comment
	PO reports all SG pressures are dropping in an uncontrolled manner or completely depressurized with MSLI failure.		
	CRS transitions to EOP-LOSC-1, Loss of Secondary Coolant.		
Note: CFSTs are in effect when transition out of TRIP-1 occurs. STA will report to control room 10 minutes after being summoned via page to monitor CFSTs.			
	RO reports MSLI failed to shut any MSIV, all MS7 and MS18 valves are shut, and action has been directed to locally shut MSIVs.		
	PO reports all SG pressures are dropping in an uncontrolled manner.		
	CRS transitions to EOP-LOSC-2, Multiple Steam Generator Depressurization.		
	PO reports all valves in table A are shut except for all MSIVs.		
	CRS dispatches operators, if not previously performed, to locally shut valves in Table A one loop at a time.		
	CRS determines 23 AFW pump in service, and either stops 23 AFW pump or leaves it in service.		
Note: 23 AFW pump may be left in			

Evaluator/Instructor Activity	Expected Plant/Student Response	SBT LOG	Comment
service if it is determined it is necessary to maintain minimum AFW flow to keep SG tubes wet.			
	IF required, PO lowers 23 AFW pump speed to minimum, trips then stops 23 AFW pump.		
	IF required, CRS dispatches operator to shut 21 and 23MS45 steam supplies to 23 AFW pump.		
	RO reports RCS cooldown rate is >100°F / hr.		
	PO reduces AFW flow to each SG to no less than 1.0E4 lbm / hr.		
CT#2 (CT-33) Control the AFW rate to not less than 1E4 lbm/hr to minimize the cooldown rate. **SAT_____ UNSAT_____**			
	When STA reports RED path on Heat Sink, CRS transitions to FRHS-1, verifies operator action was cause of RED path, and returns to procedure in effect.		
	RO reports RCS Thots are dropping.		
	RO reports RCS pressure, and reports a cooldown is in progress or RCPs are secured.		
	RO reports both PZR PORVs are closed.		
	RO reports 2PR6 is shut, and 2PR7 is open.		

Evaluator/Instructor Activity	Expected Plant/Student Response	SBT LOG	Comment
	RO resets SI, Phase A, and Phase B.		
	RO opens 21 and 22CA330.		
	PO reports all SECs are reset.		
	PO reports all 230V Control Centers reset.		
	RO resets SGBD Sample Isolation Bypass, and opens 21-24SS94s.		
	CRS directs Chemistry to sample SGs for activity and boron.		
	PO reports no indication of any SGTR.		
	RO reports RHR pumps running with suction from RWST, and no discharge flow.		
	RO reports RCS pressure is stable or rising, and stops both RHR pumps..		
	RO reports both CS pumps are running.		
	RO reports containment pressure.		
	If/when containment pressure is <13 psig, **then** RO: • Resets Spray actuation • Stops both CS pumps • Shuts 21 and 22CS2 CS pump discharge valves		

Evaluator/Instructor Activity	Expected Plant/Student Response	SBT LOG	Comment
	RO reports RWST lo level alarm has not actuated.		
	RO reports RCS Thots status. IF <375, PO removes lockouts for 21-24SJ54's, and RO shuts 21-24SJ54's.		
	RO reports subcooling is >0°F.		
	RO reports RCS pressure and trend.		
	RO reports RCS pressure is stable or rising and: • Realigns charging pump through normal charging line and adjusts 2CV55 to maintain PZR level >33%. • Reports PZR level trend.		
Simulator Operator: When Step 20 is complete then shut 23MS167 by deleting **MALF MS0092G** and ensuring 23MS167 shuts.			
	PO reports indication of 23MS167 being shut.		
	CRS determines pressure rise in 23 SG requires transition to LOSC-1.		
	CRS transitions to LOSC-1.		
Terminate the scenario when the transition to LOSC-1 is announced.			

VI. SCENARIO REFERENCES

A. Alarm Response Procedures (Various)

B. Technical Specifications

C. Emergency Plan (ECG)

D. OP-AA-101-111-1003, Use of Procedures

E. S2.OP-IO.ZZ-0003, Hot Standby to Minimum Load

F. S2.OP-AB.PZR-0001, Pressurizer Pressure Malfunction

G. S2.OP-AB.ROD-0002, Dropped Rod

H. S2.OP-AB.STM-0001, Excessive Steam Flow

I. 2-EOP-TRIP-1, Reactor Trip or Safety Injection

J. 2-EOP-LOSC-1 Loss of Secondary Coolant

K. 2-EOP-LOSC-2 Multiple Steam Generator Depressurization

ATTACHMENT 1
UNIT TWO PLANT STATUS
TODAY

MODE: 1 POWER: $1x10^{-8}$ A RCS BORON: 1711 MWe 0

SHUTDOWN SAFETY SYSTEM STATUS (5, 6 & DEFUELED):

NA

REACTIVITY PARAMETERS

Control Bank D at 130 steps withdrawn.
Core burnup 500 EFPH.

MOST LIMITING LCO AND DATE/TIME OF EXPIRATION:

3.6.2.3 for 25 CFCU action a expires in 165 hours.

EVOLUTIONS/PROCEDURES/SURVEILLANCES IN PROGRESS:

S2.OP-IO.ZZ-0003, Hot Standby to Minimum Load complete up to Section 4.3, Power Operation.

ABNORMAL PLANT CONFIGURATIONS:

CONTROL ROOM:

Unit 1 and Hope Creek at 100% power.

PRIMARY:

21 Charging pump in service
23 Charging pump C/T for emergent repair, expected return in 2 hours
25 CFCU C/T for breaker replacement

SECONDARY:

21 SGFP in service
22 SGFP O/S with steam supply valves shut

RADWASTE:

No discharges in progress

CIRCULATING WATER/SERVICE WATER:

ATTACHMENT 2

SIMULATOR READY FOR TRAINING CHECKLIST

____ 1. Verify simulator is in "TRAIN" Load

____ 2. Simulator is in RUN

____ 3. Overhead Annunciator Horns ON

____ 4. All required computer terminals in operation

____ 5. Simulator clocks synchronized

____ 6. All tagged equipment properly secured and documented

____ 7. TSAS Status Board up-to-date

____ 8. Shift manning sheet available

____ 9. Procedures in progress open and signed-off to proper step

____ 10. All OHA lamps operating (OHA Test) and burned out lamps replaced

____ 11. Required chart recorders advanced and ON (proper paper installed)

____ 12. All printers have adequate paper AND functional ribbon

13. Required procedures clean

____ 14. Multiple color procedure pens available

____ 15. Required keys available

____ 16. Simulator cleared of unauthorized material/personnel

____ 17. All charts advanced to clean traces and chart recorders are on.

____ 18. Rod step counters correct (channel check) and reset as necessary

____ 19. Exam security set for simulator

____ 20. Ensure a current RCS Leak Rate Worksheet is placed by Aux Alarm Typewriter

____ with Baseline Data filled out

____ 21. Shift logs available if required

____ 22. Recording Media available (if applicable)

____ 23. Ensure ECG classification is correct

____ 24. Reference verification performed with required documents available

____ 25. Verify phones disconnected from plant after drill.

____ 26. Verify EGC paperwork is marked "Training Use Only" and is current revision.

____ 27. Ensure sufficient copies of ECG paperwork are available.

ATTACHMENT 3

CRITICAL TASK METHODOLOGY

In reviewing each proposed CT, the examination team assesses the task to ensure, that it is essential to safety. A task is essential to safety if, in the judgment of the examination team, the improper performance or omission of this task by a licensee will result in direct adverse consequences or in significant degradation in the mitigative capability of the plant.
The examination team determines if an automatically actuated plant system would have been required to mitigate the consequences of an individual's incorrect performance. If incorrect performance of a task by an individual necessitates the crew taking compensatory action that would complicate the event mitigation strategy, the task is safety significant.

I. Examples of CTs involving essential safety actions include those for which operation or correct performance prevents...

- degradation of any barrier to fission product release
- degraded emergency core cooling system (ECCS) or emergency power capacity
- a violation of a safety limit
- a violation of the facility license condition
- incorrect reactivity control (such as failure to initiate Emergency Boration or Standby Liquid Control, or manually insert control rods)
- a significant reduction of safety margin beyond that irreparably introduced by the scenario

II. Examples of CTs involving essential safety actions include those for which a crew demonstrates the ability to...

- effectively direct or manipulate engineered safety feature (ESF) controls that would prevent any condition described in the previous paragraph.
- recognize a failure or an incorrect automatic actuation of an ESF system or component.
- take one or more actions that would prevent a challenge to plant safety.
- prevent inappropriate actions that create a challenge to plant safety (such as an unintentional Reactor Protection System (RPS) or ESF actuation.

ATTACHMENT 4

SIMULATOR SCENARIO REVIEW CHECKLIST

SCENARIO IDENTIFIER: 15-01 NRC ESG-3 REVIEWER: C Recchione

Initials		Qualitative Attributes
CR	1.	The scenario has clearly stated objectives in the scenario.
CR	2.	The initial conditions are realistic, in that some equipment and/or instrumentation may be out of service, but it does not cue crew into expected events.
CR	3.	The scenario consists mostly of related events.
CR	4.	Each event description consists of: • the point in the scenario when it is to be initiated • the malfunction(s) that are entered to initiate the event • the symptoms/cues that will be visible to the crew • the expected operator actions (by shift position) • the event termination point
CR	5.	No more than one non-mechanistic failure (e.g., pipe break) is incorporated into the scenario without a credible preceding incident such as a seismic event.
CR	6.	The events are valid with regard to physics and thermodynamics.
CR	7.	Sequencing/timing of events is reasonable, and allows for the examination team to obtain complete evaluation results commensurate with the scenario objectives.
CR	8.	The simulator modeling is not altered.
CR	9.	All crew competencies can be evaluated.
CR	10.	The scenario has been validated.
CR	11.	If the sampling plan indicates that the scenario was used for training during the requalification cycle, evaluate the need to modify or replace the scenario.
CR	12.	ESG-PSA Evaluation Form is completed for the scenario at the applicable facility.

ATTACHMENT 4

SIMULATOR SCENARIO REVIEW CHECKLIST

Initial		Quantitative Attributes (as per ES-301-4, and ES-301 Section D.5.d)
GG	2	Malfunctions after EOP entry: 1-2
GG	3	Abnormal Events: 2-4
GG	1	Major Transients: 1-2
GG	2	EOPs entered/requiring substantive actions: 1-2
GG	0	EOP contingencies requiring substantive actions: 0-2
GG	2	EOP based Critical tasks: 2-3

COMMENTS:

ATTACHMENT 5

ESG CRITICAL TASKS

15-01 NRC ESG-3

CT#1 (CT-3) Manually actuate at least one Containment Spray pump before an extreme challenge develops to the containment CSF.

Basis: Failure to manually actuate the minimum required complement of containment cooling equipment under the postulated conditions demonstrates the inability of the crew to "recognize a failure or an incorrect automatic actuation of an ESF system or component."

CT#2 (CT-33) Control the AFW rate to not less than 1E4 lbm/hr to minimize the cooldown rate.

Basis: Failure to control the AFW flow rate to the SGs leads to an unnecessary and avoidable severe challenge to the integrity CSF. Also, failure to perform the critical task increases the challenges to the subcriticality and the containment CSFs beyond those irreparably introduced by the postulated plant conditions. Thus, failure to perform the critical task constitutes a demonstrated inability by the crew to "take one or more actions that would prevent a challenge to plant safety." It also fails to prevent "a significant reduction of safety margin beyond that irreparably introduced by the scenario."

Note to Evaluator: CT numbers in parentheses are the corresponding Westinghouse ERG Rev. 2- based Critical Tasks procedure WCAP-17711-NP

ATTACHMENT 6

ESG-PRA RELATIONSHIP EVALUATION

EVENTS LEADING TO CORE DAMAGE

Y/N	Event
N	TRANSIENTS with PCS Unavailable
N	Steam Generator Tube Rupture
N	Loss of Offsite Power
N	Loss of Switchgear and Pen Area Ventilation
N	LOCA
N	Loss of Service Water
N	Loss of CCW
N	Loss of Control Air
N	Station Black Out

COMPONENT/TRAIN/SYSTEM UNAVAILABILITY THAT INCREASES CORE DAMAGE FREQUENCY

Y/N	COMPONENT, SYSTEM, OR TRAIN
N	Containment Sump Strainers
N	SSWS Valves to Turbine Generator Area
N	RHR Suction Line valves from Hot Leg
N	CVCS Letdown line Control and Isolation Valves
N	Gas Turbine
N	Any Diesel Generator
Y	Auxiliary Feed Pump
N	SBO Air Compressor

OPERATOR ACTIONS IMPORTANT IN PREVENTING CORE DAMAGE

Y/N	OPERATOR ACTION
N	Restore AC power during SBO
N	Connect to gas turbine
N	Trip Reactor and RCPs after loss of component cooling system
N	Re-align RHR system for re-circulation
N	Un-isolate the available CCW Heat Exchanger
N	Isolate the CVCS letdown path and transfer charging suction to RWST
N	Cooldown the RCS and depressurize the system
N	Isolate the affected Steam Generator that has the tube rupture(s)
N	Early depressurize the RCS
N	Initiate feed and bleed

Complete this evaluation form for each ESG.